Sprint 3: System Architecture

TEMA: A Qualitative Analysis Application

Alejandro Ciuba (PO)
Milan Knezevic (Dev)
Vaibhav Singh (Dev)
Daniel Yates (SM)

1. Revised Analysis Object Model

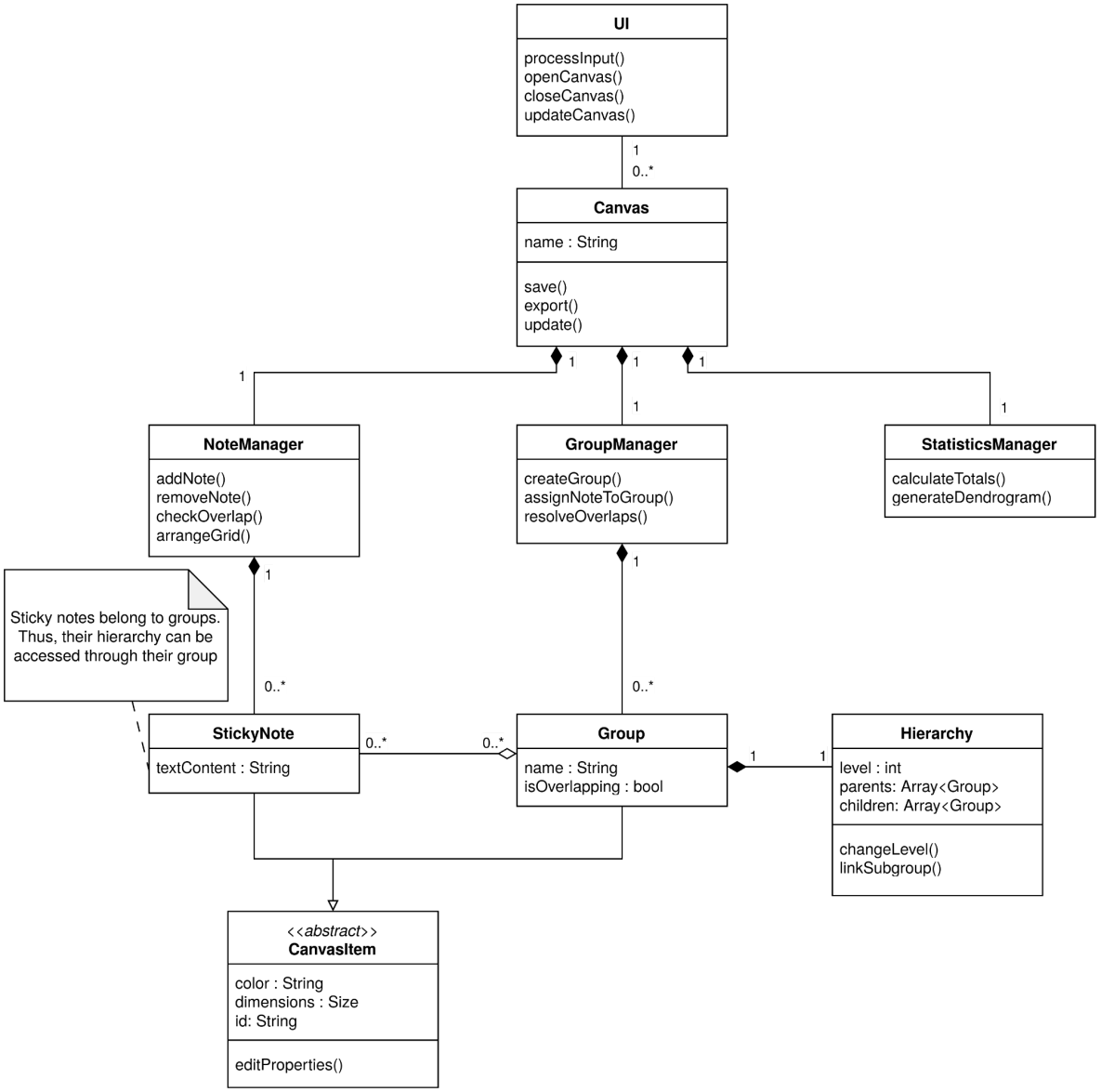

Figure 1: Revised AOM Diagram with better breakdown of the Canvas. We do not model the Google Doc/Sheet extension as that is a secondary goal to the main application.

2. Design goals

Design Goal	Measurement/Metric	Justification
Usability, Accessibility	Notes can be edited in ≤ 2 clicks	Reducing clicks for editing will allow users to easily edit any mistakes or changes they want to make, improving the user experience.
Fault Tolerance, Response Time	Customizable Autosave	Having an autosave feature which users can customize to save at different time intervals gives them both reassurance that data loss can be minimized and that any performance hits from saving can be reduced if needed.
UI Responsiveness, Response Time, Utility	No stutter even with > 200 Sticky Notes	Qualitative analysis frequently deals with large quantities of sticky notes, thus our system needs to handle large quantities of sticky note objects to be useful.

3. Subsystem Decomposition

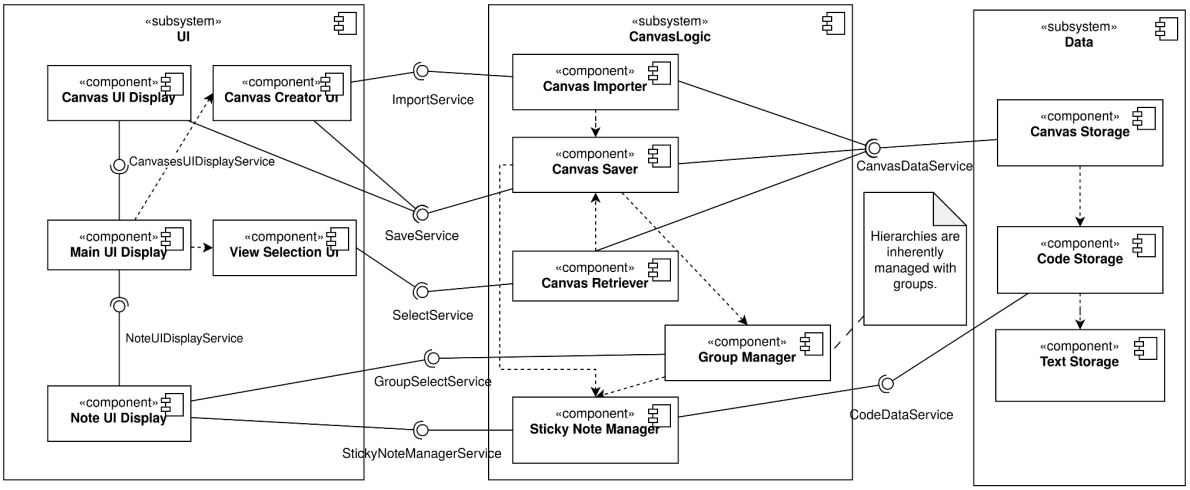

Figure 2: UML component diagram

4. Architectural Reasoning & Design Decisions

Layered architecture with different components

Why does this architectural style fit your system?

- The layered and component-based architectural style is appropriate for the system because the application has several responsibilities that are well separated, including the UI, the application logic, local data storage, and external integration with Google Doc/Sheets.

- A layered structure helps in the separation of these responsibilities, and the component-based structure helps in the modularity of major parts of the system, including canvas and notes management, visualization and export.

How does it support your design goals?

- This architecture supports the major design goals of TEMA because it organizes the system in such a way that its usability, reliability, and responsiveness are all improved. For instance, separating the interface from the logic helps keep things simple and easy to scale.

- The use of a dedicated local storage system helps improve reliability because canvases can be safely stored and restored. The organization helps improve responsiveness because of its ability to handle many notes independently.

What key design decisions did your team make?

- The user interface was separated from application logic so that canvas interaction does not depend on logic such as note/group editing, and we can change the UI without affecting the core thematic analysis behavior.

- Core note, grouping, and hierarchy operations were placed in dedicated logic components because these are the main domain functions of the system and must remain consistent across the application.

- Persistence was handled through a local storage subsystem so that user data can be saved and restored reliably while remaining on the local machine. We do not plan to store data on any server to simplify storage security concerns.

- Google Docs and Google Sheets integration was isolated in a separate external subsystem so the system can support interoperability without making the core application dependent on third-party services.

- Visualization and export features were separated from editing logic so outputs such as statistics, dendrograms, JSON, CSV, and PNG exports can be extended without changing the main workflow.

Reflections

Alejandro Ciuba: Revised and finalized AOM and Subsystem Decomposition UML Diagram.
Milan Knezevic: Drafted Architectural Reasoning and Design Decisions.
Vaibhav Singh: Designed revised *Analysis Object Model*. Drafted Design goals.
Daniel Yates: Designed Subsystem Decomposition UML Diagram.